



Concentration and Wages in Retailing

Timothy J. Richards¹ · Ujjwol Paudel² · Keenan Marchesi³

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Abstract

Antitrust policy in the U.S. now explicitly includes labor-market outcomes as measures of interest when considering the potential anticompetitive effects of mergers or acquisitions. Concentration in the food retailing industry is of particular concern due to several recent high-profile mergers, and a troubling increase in concentration at the national and local levels. We study this problem using both causal reduced-form models and a structural model of search, match, and bargaining. Our reduced-form models show no relationship between concentration and wages, but our structural model finds that concentration is associated with substantial wage suppression.

Keywords Concentration · Food price inflation · Hirschman-Herfindahl Index · Imperfect competition · Retail wages

JEL Classifications D43 · J42 · L13

1 Introduction

In 2021, the President’s Executive Order on Promoting Competition in the U.S. Economy explicitly expanded the focus of antitrust policy from prices and other consumer-centric welfare measures – variety of choices, the pace of innovation, and

✉ Timothy J. Richards
trichards@asu.edu

Ujjwol Paudel
upaudel@ttu.edu

Keenan Marchesi
kmarchesi@pm.me

¹ Morrison School of Agribusiness, W. P. Carey School of Business, Arizona State University, 7231 E Sonoran Arroyo Mall, Mesa, AZ 85212, US

² Department of Agricultural and Applied Economics, Texas Tech University, Box 42132, Lubbock, TX 79409, US

³ Economic Research Service, Washington, D.C., US

quality of service, for example – to include labor-market outcomes (POTUS 2021). In the retail food industry, public concern with regard to concentration among food retailers now extends beyond the effect of reduced competition on the price of goods and includes wages, employment, and labor-market mobility (MacDonald et al., 2023). More generally, the declining value-share of labor and rising income inequality serves as motivation for an emerging literature that examines the role of imperfect competition in labor markets from a macroeconomic (Autor et al., 2020; Berger et al., 2024; De Loecker et al., 2020) and microeconomic perspective (Arnold, 2021; Azar et al., 2020, 2022; Benmelech et al., 2022; Qiu & Sojourner, 2023; Rinz, 2022; Yeh et al., 2022). Perhaps not surprisingly, researchers generally find a negative association between employer concentration and the wages that are paid to employees, and interpret their findings as evidence of monopsonistic wage suppression.

However, none of the research on this question focuses on the food retail industry, despite the significant media attention that has been paid to supermarket concentration and its high-profile and ubiquitous role in the US economy (POTUS 2021; FTC 2024). Moreover, most of this literature uses a “neo-structure-conduct-performance” (N-SCP) approach that seeks to draw a straight line between concentration and wage outcomes, despite criticisms that such models are not valid (Berry et al., 2019; Miller et al., 2022; Syverson, 2019).

In this paper, we estimate the relationship between industry concentration and labor-market outcomes in the retail grocery industry while disentangling the underlying bargaining-power effect of concentration.

Estimating the effect of market structure on equilibrium outcomes should also reflect the fact that labor markets differ from product markets in fundamental ways: Whereas product markets are relatively easy to define based on how consumers substitute among products with similar end-uses, over relatively small geographic areas in the case of retail food products, labor is much more fungible than consumer products. Geographically, labor markets are often defined in terms of industry or occupation and commuting zone combinations (Arnold, 2021; Azar et al., 2020, 2022; Benmelech et al., 2022; Qiu & Sojourner, 2023; Rinz, 2022; Yeh et al., 2022). Focusing on industries, however, embodies a strong assumption that workers, and their skills, are not fungible across industries. In contrast, our approach in this paper recognizes that labor markets consist of individuals in occupations and not individuals in industries.

We begin by attempting to replicate the recent literature that estimates the causal effect of changes in concentration on wages in general settings to an industry that assumed particular prominence in the fall of 2024 during the federal government’s antitrust case against the proposed Kroger-Albertsons supermarket merger. Finding no effect on wages when we use generally-accepted methods, we then extend the usual approach with a structural model of labor bargaining power.

With a structural approach that is grounded in macroeconomic labor models of search, match and bargaining (Flinn, 2006), we find that concentration explains 37% to 40% of the deviation between observed wages in the grocery industry and the marginal value of equilibrium employer-employee matches. Although our treatment-effect models find no consistent relationship between retail concentration and wages, our structural model finds that concentration improves retailers’ bargaining power

substantially. For example, if wages were truly competitive, employees would earn the full amount of employment surplus that is available from a typical match: \$40.28 in our preferred model. Instead, they earn an average wage of : \$49.32% of the competitive level.

We attribute this markdown to monopsony-induced improvement in bargaining power; but we condition our findings with the caveat that deviations between competitive and observed wages in a search, match, and bargaining framework are also due to search frictions, transactions costs, and/or factors that may not reflect the exercise of market power over workers (Berger et al., 2024). Our findings are important as they provide the first estimates of the effect of concentration in the retail grocery industry on worker wages, and likely welfare outcomes that are due to rising concentration (MacDonald et al., 2023).

The rest of the paper is structured as follows: In the next section, we provide some background on the importance of antitrust policy in the retail food industry, wage setting in labor markets more generally, and how we interpret our contribution to the literature. In the third section, we describe our empirical models, first a reduced-form approach to estimate the causal effect of retail concentration on wages, and then a structural framework to estimate how concentration affects bargaining power in labor markets. In the fourth section, we describe the data that we use in both our treatment-effect and structural models and provide some model-free evidence of the relationship between concentration and wages. We follow in the fifth section by presenting the results from both our reduced-form and our structural models, and we offer some policy guidance that follows from our findings. We conclude in the sixth section and offer some suggestions for future research in this area.

2 Background

In this section, we first establish the emerging importance of worker considerations for the application of federal antitrust policy, and then discuss more specific issues with regard to the econometric identification of market power in the labor market.

The growing importance of labor market outcomes for the conduct of antitrust policy is clear from both guidance from federal officials, and the actions of firms that seek to merge with others. First, the Executive Order on Promoting Competition in the U.S. Economy in 2021 made the concern for labor explicit in revising the goals of antitrust policy (POTUS 2021). In expressing the goals of federal antitrust policy, the EO states "...to combat the excessive concentration that...has increased the power of corporate employers, making it harder for workers to bargain for higher wages and better work conditions..." (POTUS 2021), which is the first time that concerns for the labor market appear as primary concerns in policy communications; this presumably reflects a change in Federal Trade Commission policy goals as well.

Second, in announcing a planned divestiture, pursuant to a potential resolution of federal antitrust concerns regarding their proposed merger with Albertson's, officials at Kroger, Inc. explained that the buyer of the divested stores "...commits to honoring all collective bargaining agreements which include industry-leading benefits, retaining frontline associates and further investing for growth..." (Pnewswire,

2023), which again anticipates concerns with regard to the labor-market implications for their proposed merger. From the Federal Trade Commission side, the Commission's focus on labor market outcomes is clear as it argued that "...Kroger's proposed acquisition of Albertsons would immediately erase aggressive competition for workers, threatening the ability of employees to secure higher wages, better benefits, and improved working conditions..." in opposing the merger (FTC 2024).

More generally, labor markets are now more important than perhaps ever. With unemployment rates recently near to what is commonly regarded as the "natural rate" (4.31%, FRED 2025) labor markets are historically tight, with implications throughout the economy, from shortages of workers across a broad swath of industries to record-breaking strike activity from newly empowered worker groups. In short, labor concerns are now critically important to a range of federal policy decisions.

Modern antitrust considerations reflect the social and local-market impact of consolidation of legislators in the rapidly-transforming 1920s and 1930s. Indeed, anti-chain-store activists in the 1930s sought, and ultimately received, legislation that targeted the spread of the Great Atlantic and Pacific Tea Company (Purnell, 2006), and similar arguments followed Walmart's growth in the 1980s - 2000s (Basker, 2005, 2007a; Dube et al., 2007; Neumark et al., 2008). Our analysis of the labor-market effects of food retail mergers and acquisitions will follow the literature with regard to Walmart's effect on labor market outcomes during its expansion phase in the 1990s through the current century. Whether dynamics among and within retail chains lead, follow, or cause economic decline is an empirical question. In this study, we will examine whether retail mergers in the U.S. affects employment, wages, and the total welfare of employees.

Concentration in the food retailing sector – at least when measured as product-market concentration – has been rising for many years (MacDonald et al., 2023). Figure 1 shows the path of retail product-market concentration from 2004 to 2022, and a similar measure showing CZ-level labor-market concentration. This figure shows that both measures have been rising rapidly over the past 20 years, but labor-market concentration remains relatively low. Clearly, casual empiricism from simple observation of concentration rates in either product or labor markets cannot lead to solid conclusions with regard to the potential welfare implications for workers. More careful empirical analysis is necessary, and fortunately researchers in the labor economics literature have made great strides in this direction.

We first apply recent innovations in identifying the causal impact of concentration on wages to data from the food retailing industry. Our approach addresses the four key econometric issues in identifying the relationship between concentration and labor market outcomes outlined above: (1) market definition; (2) concentration endogeneity; (3) the confounding effect of product-market concentration; and (4) the structural effect of concentration on bargaining power.

First, most research in this area defines labor markets on an industry-geography basis with the use of some objective definition of an industry to control for the type of job (NAICS or SIC codes) and Commuting Zones (CZs) to define the geographic

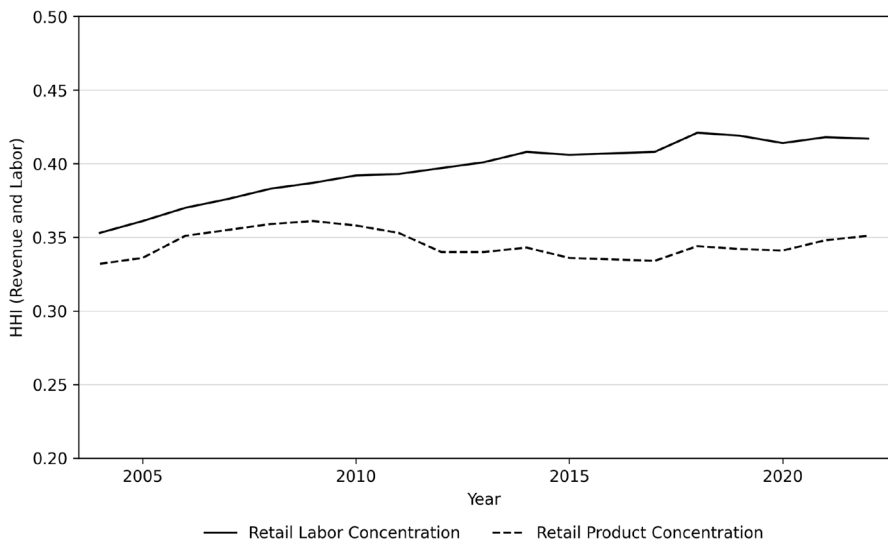


Fig. 1 Product Market and Labor Market Concentration in U.S. Food Retailing, 2004–2022. *Note:* Data are from NielsenIQ TDLinx, 2004–2022, sample of retail grocery-industry firms (NAICS=445), at the commuting-zone level. Commuting zones are defined with the use of data from ERS-USDA

extent of the market (Azar et al., 2020, 2022; Rinz, 2022).¹ However, this labor-market definition implicitly treats all jobs in a particular industry the same, whereas we show that jobs tend to be fungible across industries as skills tend to be job-specific and not industry specific. Arnold (2021), for example, uses Longitudinal Employer-Household Dynamics (LEHD) datasets from the U.S. Census to define the share of industry-level job-to-job transitions and calculates an index of substitutability among jobs in the U.S. at the CZ level. We follow his approach and use data from the CPS and ACS surveys in order to measure concentration at the job-level instead of at the industry-level. Adopting this approach is important for our focal industry because workers in the retail food industry are not likely to have industry-specific skills like teachers (Ransom & Sims, 2010) or nurses (Staiger et al., 2010), for example. We find that industry-level wage suppression does not exist at the occupation level.

Second, we use M&A transactions, or the change in concentration in a particular market, in order to address the endogeneity of observed concentration levels (Berry et al., 2019; Miller et al., 2022; Syverson, 2019). Specifically, we use changes in ownership in the TDLinx data that will be described below to identify mergers or acquisitions within the supermarket NAICS definition (4451) and changes in employment and wages over the sample period 2004–2022.² Establishment identi-

¹ “Commuting Zones” are defined by the U.S. Census Bureau as contiguous regions within defined labor market areas (LMAs) and in which workers appear to commute frequently, and in large numbers. They are defined using hierarchical cluster analysis, so are fact-based measures of revealed commuting behavior, and are not simply political boundaries as are counties or states.

² We examine the sensitivity of our findings to this NAICS definition by expanding our analysis to the 4523 code, which includes general merchandisers and club stores, and find that our results do not change qualitatively.

fiers describe individual locations of a business, and specify exact locations and the nature of the business. Enterprise identifiers, on the other hand, provide a unique code for the owner of the establishment: If the establishment identifier stays constant – say the retail banner remains “Grocer A” at a specific location in the Phoenix metro area – but the enterprise code changes, then we tag this as a M&A observation. This approach also addresses the empirical weaknesses of using continuous concentration measures over time because M&A transactions directly affect changes in concentration that are likely to change the actual exercise of market power.

Third, like others in this literature, we estimate the effect of changes in retail grocery concentration on wages by instrumenting for endogenous concentration by using M&A transactions to separate concentration from changes in concentration (Miller et al., 2022).³ We also follow the recent literature (Azar et al., 2022; Qiu & Sojourner, 2023) by using a leave-one-out measure of the inverse number of employers in CZs other than the focal CZ as an alternative instrument. By controlling for a complete set of geography level control variables, including CZ-level economic trends, our identification strategy addresses concerns with regard to the potential endogeneity of merger activity as in Hosken et al. (2018), and Arnold (2021).⁴ Our estimates that use either of these identification strategies are not qualitatively different, so we are confident in the robustness of our treatment-effect estimates.

By following the state-of-the art in this literature as closely as possible, we are ultimately unable to replicate the strong, negative relationships between industry concentration and wages in the food retailing industry. However, we then consider whether our failure to replicate more general findings is not a result of econometric identification, but rather a more fundamental flaw in thinking about labor markets from a structure-conduct-performance perspective in the first place. Instead, Card (2022) emphasizes that labor market outcomes are not necessarily the result of “market power” outcomes in a traditional sense, but individual (or group) contract negotiations that reflect perhaps more nuanced considerations of job preference, search friction, and bargaining power.

We apply this insight by using a structural search, match, and bargaining approach to estimate the underlying relationship between concentration and bargaining power. Berger et al. (2024) emphasize that monopsony wage suppression is only one potential source of imperfection in local labor markets, while search and informational frictions explain much more of the variation in local wages for similar workers in similar jobs. In fact, they find that monopsony wage suppression explains only 1/3 of the observed gap between labor marginal-value products and market wages. We use data from the ACS for occupations in the retail grocery industry in a search, match, and bargaining framework to disentangle search and matching imperfections from the effect of labor-market concentration on worker bargaining power (Dey & Flinn,

³ Despite the fact that M&A transactions occur at different points in time, our treatment-effect estimates do not suffer from the staggered-treatment OLS bias that is described in Goodman-Bacon (2021) and Callaway and Sant’Anna (2021) because our employee-employer data compares units that are always treated (employees in merged employer) with units that are never treated (employees with firms that are never merged).

⁴ Commuting-zone trends are variables that allow trend effects to vary across geographic areas, as opposed to a general trend variable that does not reflect differences in local economic patterns.

2005; Flinn, 2006; Pissarides, 2000). By allowing the share of employment surplus, or bargaining power in an equilibrium search, match, and bargaining framework, to vary with the degree of concentration in the retail-food labor market we provide the key link between market structure and conduct that is otherwise missing in the empirical literature on this topic.

Our findings contribute to the literature on imperfectly competitive labor markets (Card, 2022; Flinn, 2006; Manning, 2021; Naidu et al., 2018), specifically as it relates to the food retailing industry. Although there is a growing literature that examines the relationship between concentration and labor market outcomes in the general economy (Arnold, 2021; Azar et al., 2020, 2022; Qiu & Sojourner, 2023; Rinz, 2022; Yeh et al., 2022), there is no empirical research that focuses specifically on labor markets in the retail food industry. Insights from other sectors may not be relevant to the food retailing industry due to the prevalence of minimum-wage jobs in food retailing, the ubiquity of food retailers in nearly every county of the US, and the importance of food retailing to consumer welfare.

We also contribute to the concentration-and-wages literature by providing a key link between concentration and bargaining power. Although measures of market structure – typically the Herfindahl-Hirschman Index (HHI) – play a prominent role in antitrust guidance from the Federal Trade Commission and the Antitrust Division of the Department of Justice, they remain proxy measures for the likely use of market power and do not provide direct evidence of either imperfectly competitive pricing or wage-setting. In this paper, we demonstrate the weakness in using concentration measures in reduced-form models of wage determination, and provide what we think is a more structurally-sound way of connecting market concentration to the actual wage setting process.

Third, we contribute to the policy discussion with respect to the importance of labor-market outcomes for antitrust evaluation. From a broader welfare perspective, it is inarguable that worker outcomes are as important, and often more important, than traditional consumer-level welfare measures (Naidu & Posner, 2022; Qiu & Sojourner, 2023). Because labor-market outcomes are critically important to industrial policy, it is important that we develop a new set of methods that are generally regarded as correct, and credible, for measuring the effect of increasing concentration on workers.

In this paper, we provide empirical evidence on the relationship between concentration and wages in the retail food industry and, more generally, develop an empirical framework for testing the structural relationship between concentration and wages that may be useful in other industries, and other settings.

3 Empirical Model

3.1 Overview

In this paper, we divide our empirical approach into two sections: We first establish credible causal effects of industry concentration on wages and, finding essentially a null effect, we then offer a second structural analysis of how concentration influences

the exercise of bargaining power in individual wage relationships. We outline each in this section.

3.2 Treatment Effect Analysis

In the first of our empirical models, we estimate instrumental variables models for the relationship between concentration and wages; we use data from TDLinx and the American Community Survey (ACS, Bureau of the Census) to measure concentration, and we use both the Quarterly Census of Employment and Wages (QCEW) and the ACS to measure wages at the local labor-market (commuting zone) level. Our approach recognizes that concentration is endogenous (Arnold, 2021; Miller et al., 2022), so our main reduced-form evidence comes from instrumental variable strategy which we discuss below.

To get a baseline set of results, we also implement the following equation where we endogenize concentration:

$$\log(y_{ct}) = \alpha \log(HHI_{ct}) + \phi_c + \phi_t + \phi_c \times t + \theta \mathbf{X}_{ct} + \varepsilon_{ct}, \quad (1)$$

where: the outcome variable y_{ct} denotes the wage (average weekly) of grocery workers in commuting zone c in year t ; HHI_{ct} is a measure of the CZ-level labor market concentration (HHI); ϕ_c are commuting zone fixed effects, ϕ_t are year fixed effects, $\phi_c \times t$ is a set of commuting zone time trends, $\mathbf{X}_{ct}$ is a vector of potentially time-varying commuting-zone attributes (population, income, average age, education level); and ε_{ct} is the error term. We estimate in logs so that our estimated treatment effects are interpreted as elasticities, and so as to remain comparable to others in this literature. In this specification, the parameter α measures the treatment effects of HHI on CZ-level wages of workers in retail grocery establishments.

Our main IV model leverages Hausman-like instrumental variables for endogenous employment-HHI levels. Specifically, Azar et al. (2022), Rinz (2022) and Qiu and Sojourner (2023) each use the average concentration for the same occupation across all other CZs in the same year (a “leave-one-out” instrumental variable (LOOIV)). This is similar to the Hausman-instrumentation strategy (Hausman, Leonard, and Zona 1994) that is frequently used in the empirical industrial organization literature, and is based on the assumption that local labor-demand shocks are independent of national labor-demand shocks that may cause all local concentration levels to move together. In order to maintain comparability with the existing literature, we present estimates that use this IV strategy as our primary results as opposed to results from the number of CZ-level mergers as an IV for HHI.⁵

As robustness checks, we estimate versions of equation (1) without commuting-zone level controls, fixed effects, and commuting-zone level trends. Further, we esti-

⁵ We use contemporaneous mergers, instead of their cumulative effect, for several reasons: A contemporaneous-merger-count instrument is preferred because it better matches the timing of the outcome variable (wages); it is strong in a first-stage IV regression ($F = 601.08$); the exclusion restriction is much more plausible with the contemporaneous measure; and it is more likely to be stable, or stationary, relative to a cumulative measure. In using a contemporaneous measure of mergers, we also follow Azar et al. (2022) and Qiu and Sojourner (2023), who use much the same line of argument that we do.

mate specification (1) with the use of the number of mergers as an IV for HHI, similar to the strategy in Arnold (2021). In this case, we find point estimates of the HHI-wage relationship that are consistently negative, but again not significantly different from zero, as in our estimates with the LOOIV (see table A1 in the appendix). Our preferred instrument also aligns with the notion that mergers are themselves endogenous (Gowrisankaran (1999); Hosken et al. (2018)). We also estimate the specification with wages that are derived from the ACS rather than from the QCEW.

As we show below, our findings from all of these model variations are qualitatively consistent, and show that there is little statistical evidence of a direct relationship between concentration in the retail labor market and wage outcomes. However, our findings do not rule out an indirect, structural relationship between concentration, bargaining power, and wages. In the next section, we describe a structural model that is able to disentangle the actual mechanism that underlies the relationship between concentration and wages.

3.3 Structural Model Analysis

In this section, we provide a brief, intuitive explanation of the core of our theoretical model, and reserve a more detailed exposition of each model element to the appendix. In this model, concentration affects employee-level outcomes through a structural framework in which bargaining and search frictions are as likely to affect wages as is bargaining-power-driven wage suppression.

We apply a model of labor market search, matching, and bargaining that was developed by Flinn (2006) to examine equilibrium employment, wage, and productivity outcomes among our sample of retail workers. Job duration, unemployment, and wages are each equilibrium outcomes in the sense that firms search optimally for workers, workers search optimally for jobs, and search occurs until the point at which the marginal benefit of additional search effort is just equal to the marginal cost of doing so (Stigler, 1961). When both firms and workers optimize, the “employment surplus,” or the amount of value created by the worker for the firm, is at a maximum.

Our framework departs from the usual “take it or leave it” assumption in the labor economics literature (Burdett & Mortensen, 1998; Eckstein & Van den Berg, 2007; Van den Berg & Ridder, 1998) by allowing workers and firms to bargain over wage outcomes. Negotiation between firms and workers, or their representative, means that the amount of employment surplus that is shared between workers and firms according to their relative bargaining power, which is exogenous to each party and is determined by individual- and firm-level attributes, is interpreted as negotiating “skill” (Muthoo, 1999; Nash, 1950).

In our model, one of the primary attributes that determines the balance of bargaining power between firms and workers is the relative number of employers and employees in each labor market, or the concentration measures developed in the previous section. In general, we expect to find that the degree of bargaining power that is possessed by retail workers will be relatively low, given the fungible nature of retail job skills relative to other, more specialized industries.

An equilibrium model of search, match, and bargaining is appropriate for the problem at hand because equilibrium models are necessary to account for many

well-understood sources of labor-market imperfection: Search frictions, transactions costs, and partial information with regard to job openings and opportunities (Berger et al., 2022, 2024). Once we control for all other possible labor-market imperfections that can lead to rents in labor-market transactions, the only other logical factor left is simply employer bargaining power.⁶ With our structural model, we allow the structure of the market for retail workers to explain the allocation of rents that are left over after accounting for all of these other factors. In this way, our structural model provides a clean test of the potential for labor market power to lead to adverse outcomes for workers.

Among all of the variations on search, match, and bargaining models in the literature (Dey & Flinn 2005; Flinn 2006; Flabbi & Moro 2012, for example), the Flinn (2006) approach is particularly useful in our context as it is specifically intended to examine bargaining in a setting where minimum wages are important. In the food retailing industry, others document the fact that fully 25% of retail workers earn at or near the minimum wage (Leung, 2021; Renkin et al., 2022); and in our broader set of retail workers from the CPS we find that nearly 14.0% earn at or near the minimum wage. Accounting for minimum-wage workers is critical as employers essentially give up some surplus to minimum-wage workers when their marginal value products are still higher than the minimum wage and the minimum lies above the wage that would otherwise represent an equilibrium (Flinn, 2006). We are the first to explore how wages – and more precisely employment-surplus allocation – depend on the relative bargaining power strength of workers and firms in concentrated, minimum-wage industries.⁷

We use this model to derive several testable hypotheses with regard to the relationship between concentration and wages, and the allocation of employment surplus, productivity, job duration, and unemployment duration. Our structural model identifies six key parameters that describe the equilibrium labor-market flow between employers and employees, and how employment surplus (the difference between the marginal value generated by each worker and their "threshold wage," or the wage that causes them to accept an offer) is divided between workers and firms according to their respective bargaining powers.

First, τ represents the equilibrium rate of job contacts, or the rate at which new jobs are created in the economy. The flow of workers into the economy through new-job creation is offset by the rate of job destruction (δ), which captures not only layoffs and firings, but also separations for any reason, both firm- and employee-determined. Employee productivity (ϕ) follows a log-normal distribution $f(\phi)$ and guides each firm's willingness to pay for a new employee, and the amount of surplus

⁶Note that bargaining power differs from monopsony power in the sense of Robinson (1969) or Manning (2003) in that we do not rely on the ability of a firm to limit the amount of labor it employs, but rather focus on how its position in the industry affects its ability to negotiate, or bargain, with workers. In a Nash (1950) sense, industry dominance is a natural and intuitive way to think of how firms gain bargaining power over workers.

⁷We are not the first to use a formal model of optimal search, match and bargaining to explain labor market outcomes: Flinn (2006) examines how minimum wage laws affect equilibrium wages and employment; Dey and Flinn (2005) consider how wages differ between those who have healthcare and those who do not; and Flabbi and Moro (2012) consider gender differences in compensation.

over employee wages (w). Employers' wage offers are assumed to be constrained by an exogenously-determined minimum wage (w_m) that varies by jurisdiction, and over time. Employees search optimally for employers, in the sense of Burdett and Mortensen (1998), and employers post wages for each job opening. Once a match is established, equilibrium wages emerge as a bargaining equilibrium that allocates the total amount of employment surplus to the employee (λ) and the firm ($1 - \lambda$).

We capture the possibility that the surplus-allocation function depends on market structure by allowing the Nash bargaining parameter $-\lambda(h)$ to be a linear function of local labor-market concentration (h): We allow the bargaining-power parameter to be a function of concentration so that: $\lambda(h) = \lambda_0 + \lambda_1 HHI_{ct}$, where HHI_{ct} is the Hirschman-Herfindahl Index (HHI) of local (commuting zone, c) labor market concentration in period t , as in our reduced-form model above. If concentration alone provides employers with sufficient market power to suppress equilibrium wages below the competitive level, holding all other sources of market imperfection constant, then $\lambda_1 < 0$. In the next section, we: provide more detail on the data that we use to estimate this model; explain how we define local labor market concentration; and provide some model-free evidence on the relationship between concentration and labor-market outcomes.

4 Data and Identification

Our primary data sources for this paper are: the TDLinX retailer-attributed data set from NielsenIQ; the Quarterly Census of Employment and Wages (QCEW, Bureau of Labor Statistics, U.S. Department of Labor); the Current Population Survey (CPS, Bureau of Census); and the American Community Survey (ACS, Bureau of Census). We access the latter two data sets through iPUMS (Ruggles et al., 2023). We explain how we use each data set in this section; but, to summarize, we use: TDLinX to measure the number of workers who are employed by each firm in each market, which we define as a commuting zone; the QCEW to measure weekly wages for workers in the food retailing industry (NAICS = 445); and the other data sets to measure the number of workers in retailing-related occupations in each market. We also use the ACS data exclusively to estimate the equilibrium model of search, match, and bargaining.

Our store-level retail-employment data are from NielsenIQ's TDLinX Store Characteristics Dataset, which provides detailed estimates of: annual store-level store volume (which is measured in terms of dollar sales); the number of employees; and a variety of other store-attribute variables that permit a granular and detailed description of each establishment. Most important: We know the exact location of each store, its format classification, its ownership group, and whether it changes ownership from one year to the next.⁸ TDLinX is a census in the sense that it covers every store in the U.S. in the supermarket, mass merchandiser, drug store, dollar store, and convenience store channels that may be of interest to potential market entrants. In this regard, NielsenIQ intends for TDLinX to be used as a strategic information source,

⁸ It is important to note that TDLinX tracks the building itself, so we ensure that we track true openings and closures and not merely changes in ownership at the same location.

which allows firms to obtain comprehensive market intelligence for every market in the U.S. We use the information in TDLinx to infer whether or not each location was part of a merger or acquisition during the sample year.

We use the geographic information in TDLinx to associate each store with a county (FIPS code), and then use a cross-walk file from ERS-USDA to associate each county with a commuting zone.⁹ We choose a relatively long sample period – 2004 to 2022 – in order to capture as much of the rise in retail concentration as possible, and the wage changes in each retail occupation. Our geographic sample frame is the entire U.S., so our data represent workers from approximately 45,000 grocery stores, which account for over 12.9 million workers in the most recent years in the data. For our initial models, we calculate local labor-market concentration by calculating the share of retail grocery workers who are employed by each ownership group, and then calculate the Hirschman-Herfindahl Index (HHI) of local labor market concentration by squaring and summing the shares over all employers in the same commuting zone. With this measure of concentration, therefore, the denominator in our share calculations is the total employment by all food retailers in the local market.

We measure wages in the grocery industry by using the QCEW data set. The QCEW is an administrative data set that is collected by the U.S. Department of Labor. It is particularly useful for this purpose as QCEW is a census, since all employers that participate in state-level unemployment insurance programs are required to submit data on compensation and employment to their respective state agencies, who then report their data on employment and wages (and, implicitly, the number of establishments based on the number of firms that report) to the BLS. In this way, the QCEW provides a consistent measure of average weekly wages over the entire sample period, and at the same level of geographic disaggregation as our TDLinx data.¹⁰ In order to focus our empirical model on the food retailing sector, we extract data for NAICS = 445 (Supermarkets and other Grocers) for every year and geography in our sample frame. We convert all nominal wages to real terms with the use of the Consumer Price Index for All Urban Consumers (CPI-U, BLS), expressed in 2004 dollars.

Our final data set consists of a total of $N = 45,180$ commuting-zone-year-quarter observations. We summarize the combined TDLinx and QCEW in Table 1.

Insights with regard to the relationship between concentration and wages are relatively clear from the combined TDLinx and QCEW data: In Figure 2, we show a summary relationship from a simple scatterplot of wages for food-retailing workers on the vertical axis and retail-labor concentration (measured using the HHI) on the horizontal axis (each dot represents a quarter/commuting zone observation). Without

⁹ In this literature, it is well-understood that the geographic extent of a labor market is appropriately described as commuting zone. The ERS-USDA data documentation describes the need for commuting zones as "...County boundaries are not always adequate confines for a local economy and often reflect political boundaries rather than an area's local economy. CZs and LMAs [Labor Market Areas] are geographic units of analysis intended to more closely reflect the local economy where people live and work. Beginning in 1980 and continuing through 2000, hierarchical cluster analysis was used along with the Census Bureau's journey to work data to group counties into these areas..." (ERS-USDA 2023). We are also careful to track changes in county boundaries that occur over the sample period.

¹⁰ The BLS reports QCEW on a FIPS (county) basis, so we match to the TDLinx data through the same county-to-commuting zone crosswalk file that we refer to above.

Table 1 Summary of Estimation Data

Variable	Source	Units	Mean	Std. Dev.	Min.	Max	N
Wage	QCEW	\$/Week	361.76	117.84	50.29	926.75	14,372
Employment	QCEW	# Workers	2,157	3,572	132	32,996	14,372
Retail Labor HHI	TDLinx	Index	0.2250	0.0740	0.0709	0.5475	14,372
Retail Output HHI	TDLinx	Index	0.2663	0.0882	0.0783	0.5465	14,372
ACS HHI	ACS	Index	0.0066	0.0151	0.0001	0.2189	14,372
Income	Census	\$/Year	52,874	12,377	23,188	117,948	14,372
Unemployment	Census	%	6.32	2.71	2.05	27.20	14,372
Population	Census	,000	1,283	1,926	104	18,700	14,372
Age	Census	Years	39.26	2.37	31.40	48.66	14,372
Employers	Census	#	15.79	8.97	4.00	66.00	14,372
LOOIV	Census	#	1.16	0.08	1.06	1.33	14,372
Number of Mergers	Census	#	20.76	34.27	0.00	517.00	14,372

All data are on a commuting-zone basis from 2004–2022 and are from either the Quarterly Census of Employment and Wages (QCEW, Bureau of Labor Statistics), American Community Survey (ACS, Census Bureau), or TDLinx (NielsenIQ). All Hirschman-Herfindahl Indices (HHI) are divided by 10,000 from the usual definition for estimation purposes

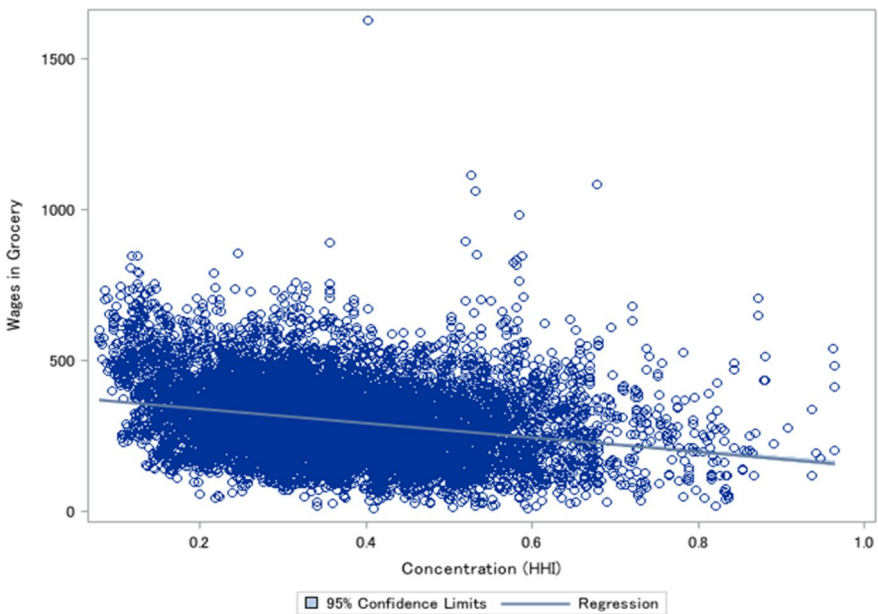


Fig. 2 Retail Labor Concentration and Wages. *Note:* Data are from NielsenIQ TDLinx and Bureau of Census (QCEW), 2004–2022, sample of retail grocery-industry firms and workers (NAICS=445), at the commuting-zone level. Commuting zones are defined with the use of data from ERS-USDA

controlling for any other variables that may affect the relationship between concentration and wages, this figure shows, roughly, what others in the empirical literature document more formally: a concentration-wage elasticity of roughly -0.05 , which is relatively small but statistically significant. Approximately at the mean of the con-

centration data ($\text{HHI} = 0.4$) this means that an increase in concentration of 0.5 (25%) would be associated with a decline in wages of 1.25% (Figure 2). Although small, for low-wage workers a reduction in income of this magnitude would be important.

Workers in the TDLinx data, however, represent only a small slice of the total number of people who could likely fill the jobs represented in the data. If the objective is to measure job market concentration and not product market concentration, then it is necessary to find a broader measure of the number of workers in retailing jobs rather than just those currently employed in the retailing industry. For example, cashiers in the food retailing industry could just as easily be cashiers in the tire-store industry, sporting-goods industry, or any other set of firms who employ cashiers.

Therefore, we broadened our search for workers in the retailing industry to the CPS (Bureau of Census) in order to determine what fraction of retail occupations were actually employed in the food retailing industry. For the same sample period, we adopted a conservative approach (as there are likely to be workers with the same skills in many other industries) and restricted our search to a sample of workers from NAICS two-digit codes 44 and 45 and examined the range of occupations across all of the sub-industries involved.

As is shown in Table 2 below, the most common retailing occupations are Retail Supervisors and Managers and Retail Salespersons, followed by Cashiers. More important, we find that of the 31, 852 CPS workers in any retailing occupation, only 1, 463 (4.6%) were employed in food retailing. Because food retailing represents only a small part of the market for retailing jobs, it is necessary to define job-market concentration using the entire market for retailing jobs, and not just those employed directly in food retailing. Although it is possible to use the CPS sample weights to generalize this insight to the entire U.S. economy, there are many commuting zones that are not represented in the relatively small sample of CPS workers. Therefore, we further broadened our search for retailing jobs to the much larger ACS data set.

The ACS is generally regarded as the definitive survey of the U.S. population for demographic, geographic, housing, and work attributes (Antman et al., 2023; Richards & Rutledge, 2023). In order to get a better picture of the totality of employment

Table 2 Top 10 Occupations in US Retailing, CPS Data, 2010–2023

Occupation	OCC 1 Code	Frequency	%	OCC 2 Code	Frequency	%
Retail Supervisors & Managers	4700	3,231	15.61	4700	3,271	15.80
Retail Salespersons	4760	2,917	14.09	4760	3,165	15.29
Cashiers	4720	1,940	9.37	4720	1,893	9.14
Not Specified	0	1,768	8.54	0	1,538	7.43
Customer Service Reps	5240	583	2.82	5240	689	3.33
Inventory Clerks	5620	423	2.04	5620	339	1.64
Freight and Stock Managers	9620	422	2.04	9620	445	2.15
Pharmacists	3050	401	1.94	3050	421	2.03
Truck Drivers	9130	399	1.93	9130	416	2.01
Stockers and Order Fillers	9645	364	1.76	9645	507	2.45

Data in OCC 1 reflect occupations for CPS - ASEC 12-month apart samples in the first year, and OCC 2 reflect occupations for the same individuals 12 months later. Top 10 total for OCC 1 is 60.12% of employees in US retailing, and for OCC 2 is 61.26% of sample employees. Ranked by Year 1 frequency. $N=20,704$ workers

in U.S. retailing, and its relationship to concentration in the food retailing industry, we drew a sample of ACS workers equivalent to the CPS sample that we described above, including workers in the same industries and occupations as in our CPS data. Over the entire 2005 – 2022 time period, our ACS sample of retail workers produced $N = 4.258$ million observations. Because ACS is still a sample, however, we then used the ACS sample weights to construct a data set of the total number of workers in each retailing occupation, on a commuting-zone basis, and merged it with the TDLinx data in order to calculate a revised set of labor-market HHI measures that represent the true share of retail workers who are employed by food-retailing firms in each commuting zone.¹¹ Figure 3 below shows the path of retail-worker concentration from the merged ACS-TDLinx data, which shows a very different picture from the TDLinx-only data in Figure 2.

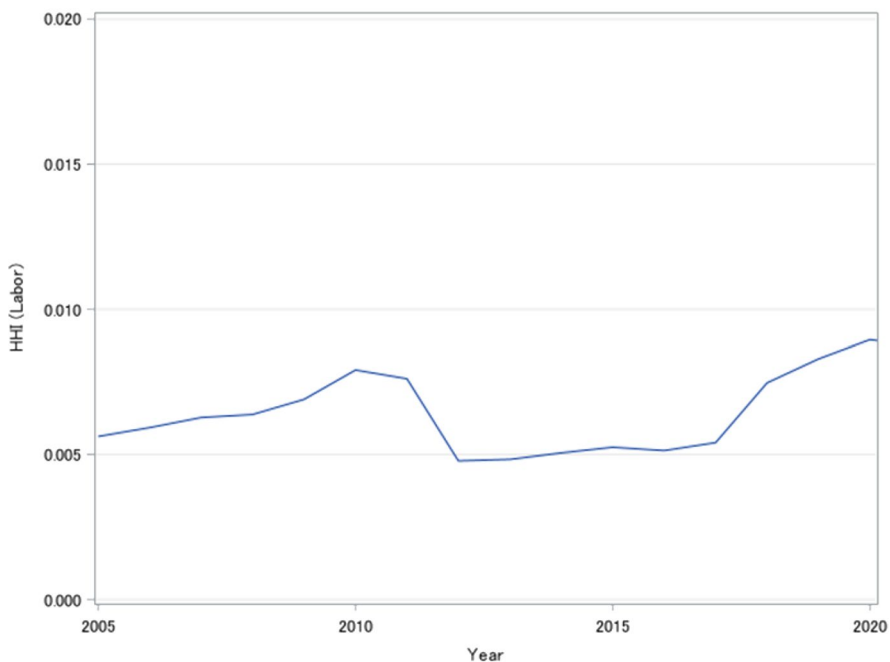


Fig. 3 Retail Labor Market Concentration: 2005 - 2022, by Commuting Zone, ACS Data. *Note:* Data are from NielsenIQ TDLinx and Bureau of Census (ACS), 2004-2022, sample of retail grocery-industry firms and workers (NAICS=445), at the commuting-zone level. Commuting zones are defined with the use of data from ERS-USDA

¹¹ This occupation-by-geography definition of a labor market is admittedly expansive, but is similar to that used in Qiu and Sojourner (2023) and others. Nimeczik (2017) uses matched employee-employer administrative data from the Austria to map flows between firms instead of within geographic regions. He finds substantial heterogeneity in who moves between jobs and shows that "...[O]n average, the spatial distance between firms within endogenous labor markets for high skilled individuals is about 1.3 times bigger than for low-skilled individuals....," which supports our notion that low- and semi-skilled workers tend to move among jobs within the same geographic area. Moreover, in our CPS-ASEC data, we find that 41.6% of cashiers are likely to move within industries, in the same commuting zone, each year.

Measured across all retailing-relevant occupations, the average ACS-HHI value is 0.006, or an order of magnitude lower than measured using food retailing workers alone. This finding is intuitive as there are simply far more retail-adjacent occupations among workers in the general population than just those represented in the TDLinx data. Yet, Figure 4 shows that the negative relationship appears to persist, despite greater skew in the distribution of the measure of concentration at the commuting zone level. Clearly, we require a more careful econometric analysis to determine whether this summary relationship is statistically significant.

We also use the ACS data for the search, match, and bargaining analysis that connects concentration to wages through a bargaining power mechanism. Search, match, and bargaining models of the sort that were developed by Pissarides (2000) and Flinn (2006) require individual-level data on hourly wages, unemployment duration (in the past 52 weeks), minimum wages in the employee's jurisdiction, and demographic and socioeconomic variables in order to control for any systematic variation in wages that is not otherwise explained by the search process, and the equilibrium that is assumed to result. Structural models, however, are more computationally difficult than are the treatment-effect models that we described above, so we sample 10% of the ACS sample described above, which is more than enough to identify the structural bargaining parameter according to the Monte Carlo experiment that is described by Flinn (2006). For the wage-estimation equation that we embed in the structural bargaining model,

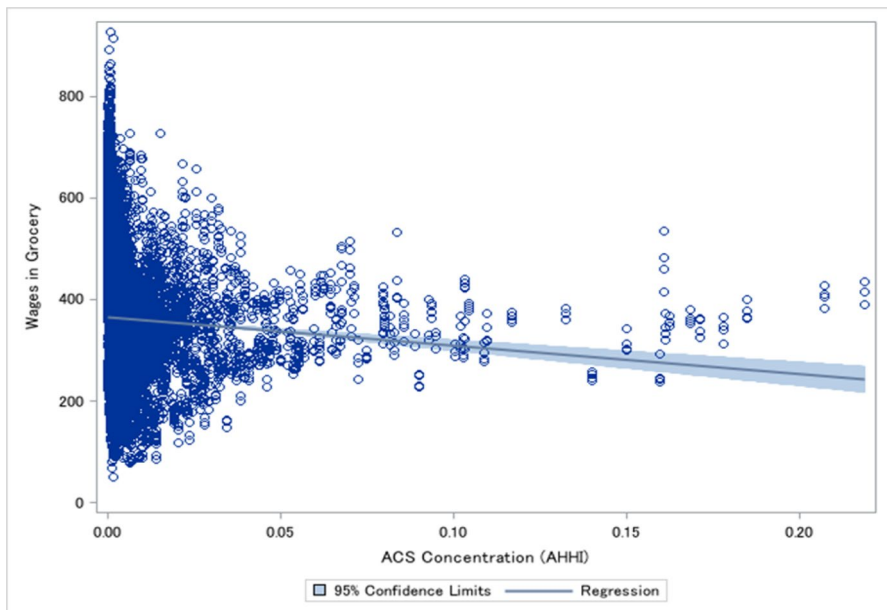


Fig. 4 Retail Labor Market Concentration and Wages: 2005 - 2022, by Commuting Zone. *Note:* Data are from NielsenIQ TDLinx and Bureau of Census (ACS), 2004-2022, sample of retail grocery-industry firms and workers (NAICS=445), at the commuting-zone level. Commuting zones are defined with the use of data from ERS-USDA

we include state, occupation, worker classification, and age, age squared, education, gender, marital and citizenship status among the other covariates.¹²

We merge the ACS data with state-level minimum wages that are from the St. Louis Federal Reserve Board (FRED 2023), and the TDLinx data above in order to allow the bargaining parameter to vary with commuting-zone level labor-market concentration values that we described above. With respect to the importance of minimum wage data, we find that some 14.0% of our retail workers earn at or near the minimum wage, so the Flinn (2006) approach is appropriate for our data.

Each of the other parameters in the structural search, match, and bargaining model is well identified in the data. First, the rate of job destruction (δ) is identified through variation in individual unemployment rates, the distribution of unemployment duration, and the distribution of wages (accepted offers) – which pins down the flow from unemployed to employed states across workers. Similarly, the rate of job contacts (τ) is also identified by the distribution of wages, and the implied rate at which new jobs are accepted by currently-unemployed workers. Third, the parameters of the productivity distribution (ϕ and σ) are identified by the equilibrium wage (w) that is offered by firms, and accepted by workers, and the mass of wages at the minimum (w_m). By accounting for cross-sectional variation in wages across workers with different demographic profiles and job attributes, variation in the observed wage distribution and unemployment across these groups helps identify heterogeneity in contact functions and reservation utilities. Fourth, the shape of the distribution of accepted offers identifies the threshold wage, or critical match value: ϕ^* .

Finally, the bargaining power parameter (λ) is identified implicitly by a combination of the slope and spread of the wage distribution and the contact rate elasticity ($\tau(w)$): A more compressed wage distribution (conditional on productivity) suggests a lower worker surplus share: If most wages are near the minimum wage, that implies low worker power. On the other hand, a right-skewed distribution suggests firms must offer higher wages to attract labor, which implies more worker bargaining power. Further, the model assumes that the contact rate (τ) rises in the posted wage, so the slope of this function tells us how responsive worker search effort is to higher wages, which in turn affects wage posting: The steeper is $\tau(w)$, the more firms need to raise wages to attract workers, which implies higher implicit worker bargaining power. We test our core hypothesis that concentration affects retail-worker wages, and their share of the equilibrium employment surplus, conditional on the identification assumptions in the search, match, and bargaining approach.

5 Results, Robustness and Discussion

In this section, we present our empirical results and robustness checks. We begin by presenting a series of reduced-form models of the relationship between labor-market concentration in an attempt to replicate others' causal results for a wider set of industries. We follow by presenting the findings from our structural model of labor market

¹²We expand our analysis to consider the impact of unionization in a similar way using a smaller sample of CPS-ORG data in a robustness check to our main results (see the appendix).

equilibrium in which concentration is allowed to affect the allocation of employment surplus between employers and employees in each market.

5.1 Reduced-Form Estimates

Table 3 shows our econometric estimates for the relationship between labor-market concentration and real wages with the use of only the TDLinx data, and a number of alternative models that are intended to examine the robustness of our maintained model. In this table, the concentration index (HHI) is defined as the sum of squared shares of each retailer relative to the total number of workers who are employed by all retailers in the same market. We use this “naive” measure of concentration to demonstrate the difference between econometric estimates that use a traditional, industry-based definition of concentration. In Table 3, we make no attempt to control for the likely endogeneity of labor-market concentration (Berry et al., 2019; Miller et al., 2022; Syverson, 2019) in order to establish a baseline set of results.

Beginning from the simplest model (Model 1) that includes only the HHI as a regressor, the results show a strong negative correlation (elasticity of -0.378) between concentration and wages. In a very general sense, these results are consistent with the recent literature on this topic (Azar et al., 2022; Qiu & Sojourner, 2023; Rinz, 2022) in that we find that as concentration rises, observed wages fall, presumably due to the monopsony suppression of wages.

However, if we add a set of minimal controls to the OLS model (Model 2, population, income, age, and unemployment rate) in order to capture the likelihood that wages depend on the relative strength of local economic activity, most of the negative effect goes away, and the point estimate of -0.043 is not significantly different

Table 3 Regression Estimates for NAICS 445: HHI and Wages, CZ Level

Dependent Variable	Log Real Weekly Wages				
	Model 1	Model 2	Model 3	Model 4	Model 5
log(HHI)	-0.3778^{***} (0.0328)	-0.0434 (0.0376)	-0.0652^{***} (0.0240)	-0.0444^* (0.0235)	-0.0382 (0.0254)
<i>Specifications:</i>					
Controls	No	Yes	Yes	Yes	Yes
CZ Fixed Effects	No	No	Yes	Yes	Yes
Year Fixed Effects	No	No	No	Yes	Yes
CZ Trends	No	No	No	No	Yes
R^2	0.1316	0.2426	0.8121	0.8164	0.8571
RSS	7,890.82	6,870.74	1,704.18	1,665.85	1,295.97

All estimates with TDLinx and QCEW data for NAICS = 445 (Grocery Retailers) on a commuting-zone basis for the years 2005–2022. HHI is defined as the Hirschman–Herfindahl Index of employer concentration in each commuting zone, which forms an “industry” definition of the relevant labor market. Model 1 consists of HHI only, Model 2 is Model 1 plus controls (population, income, age, and unemployment rate), Model 3 is Model 2 plus CZ fixed effects, Model 4 is Model 3 with year fixed effects, Model 5 is Model 4 with CZ-level trends. All standard errors are clustered at the CZ level. A single asterisk (*) indicates significance at a 10% level, ** at 5%, and *** at 1%. Dependent variable is log QCEW average weekly real wage at a commuting-zone level

from zero:¹³ Most of the observed negative relationship between concentration and wages is explained by local economic activity – and not by wage suppression. When we control for unobservable factors at the CZ level (Model 3), the point estimate is negative and statistically significant at the 1% level; once we add year fixed effects (Model 4), the estimate is similar in magnitude but weakens to significance at only the 10% level; and once we account for CZ-level trends in the most comprehensive model (Model 5), the point estimate is again similar in magnitude but is no longer statistically different from zero. In summary, the OLS evidence shows little robust support for a negative concentration-wage relationship: the apparent effect is not stable across specifications and fades once local economic conditions and trends are absorbed. But, the estimated negative relationship in Qiu and Sojourner (2023) was also very close to zero before instrumenting for the endogeneity of local-labor-market concentration.

In Table 4, therefore, we control for HHI endogeneity by instrumenting for local concentration in a manner similar to Azar et al. (2022) and Qiu and Sojourner (2023). We calculate the leave-one-out inverse number of employers in all other commuting zones – LOOIV – and use two-stage-least-squares (2SLS) to estimate the effect of instrumented-concentration on the log of average weekly real wages.¹⁴

Table 4 IV Model Estimates for NAICS 445: HHI and Wages, CZ Level

Dependent Variable	Log Real Weekly Wages				
	Model 1	Model 2	Model 3	Model 4	Model 5
log(HHI)	−0.1139*** (0.0553)	−0.0132 (0.0872)	−0.3978 (0.2269)	0.0220 (0.0531)	0.0212 (0.0435)
<i>Specifications:</i>					
Controls	No	Yes	Yes	Yes	Yes
CZ Fixed Effects	No	No	Yes	Yes	Yes
Year Fixed Effects	No	No	No	Yes	Yes
CZ Trends	No	No	No	No	Yes
R^2	0.0674	0.2422	0.8030	0.8160	0.8570
RSS	8,474.31	6,874.27	1,787.34	1,669.07	1,297.25

All estimates with TDLinx and QCEW data for NAICS = 445 (Grocery Retailers) on a commuting-zone basis for the years 2005–2022. HHI is defined as the Hirschman–Herfindahl Index of employer concentration in each commuting zone, which forms an "industry" definition of the size of the relevant labor market. Model 1 consists of HHI only, Model 2 is Model 1 plus controls (population, income, age, and unemployment rate), Model 3 is Model 2 plus CZ fixed effects, Model 4 is Model 3 with year fixed effects, Model 5 is Model 4 with CZ-level trends. IV models use a leave-one-out (commuting zone and year constant) measure of inverse employer numbers (Azar et al., 2022; Qiu & Sojourner, 2023). All standard errors are clustered at the CZ level. A single asterisk (*) indicates significance at a 10% level, ** at 5%, and *** at 1%. Dependent variable is log QCEW average weekly real wage at a commuting-zone level

¹³ As a robustness check, we included product-market concentration (revenue) as an additional regressor as in Qiu and Sojourner (2023), but our results do not change qualitatively. Because of the obvious endogeneity of product-market concentration (Miller et al., 2022), we do not include output-market concentration among our formal robustness checks.

¹⁴ The units for the LOOIV are the inverse number of employers, in native values, and are not indexed in any way.

Following Qiu and Sojourner (2023) we believe that this instrument is likely to be valid because changes in the number of employers on a national level is likely to be driven by broader economic forces that are independent of local wages, and yet highly correlated with changes in local labor-market concentration. For example, if an economic downturn has a cleansing effect (Barlevy, 2002) that increases the productivity of workers who remain on a reduced-number of jobs, there is likely to be a lower number of employers, both locally and nationally. The number of employers in other commuting zones, therefore, is likely to be highly correlated with concentration in each local market; and yet the change in the number of employers nationally does not have any logical connection to local monopsony wage suppression.¹⁵ In fact, in our first-stage IV regression, the F-statistic in a regression of only local HHI on the LOOIV produces an F-statistic of 378.4, which is far above the Andrews et al. (2019) threshold of 10.0.

Our IV estimates in Table 4 show a slightly different picture of the relationship between concentration and wages in local labor markets. In the base model (Model 1) in which we estimate the effect of concentration on wages only, with no other controls, we find a significant and negative effect of local HHI on wages. However, when we account for the same set of CZ-level controls as in the Table 3 models, we find that the negative point estimate remains in Models 2 and 3, but the estimate becomes not statistically different from zero, and turns positive and not statistically significant in our preferred specification.¹⁶ After controlling for the endogeneity of concentration, we still continue to find that occupation-based employment concentration in the retail food industry has little or no effect on wages. However, estimating this relationship with the TDLinx data accounts only for an industry-based definition of concentration and not for an occupation-based definition.

In Tables 5 and 6, therefore, we use an ACS-based measure of concentration, as we described above. As with the TDLinx analysis in Tables 3 and 4, we begin by estimating the same set of models as in Table 3; but we define concentration as the HHI that is calculated from the share of workers in the retailing industry that are employed by each store as a ratio of the total number of workers in retailing occupations. Our HHI measure in this case is likely to be a more accurate measure of the concentration of grocery retailers in the market for retail workers, rather than just the number of supermarket workers as in Tables 3 and 4.

In Table 5, we see that the reduced-form estimates of the concentration-real-wage effect is generally negative for all five specifications, which supports the findings of others in this literature (Azar et al., 2022; Qiu & Sojourner, 2023; Rinz, 2022). In

¹⁵We use inverse employer counts from other CZs, and not HHI, because the HHI depends on heterogeneity in the numerator of the HHI calculation, or heterogeneity in employers across different markets (commuting zones in our case). Instead, the inverse employer count represents a clear, mechanical, lower-bound to HHI that captures national trends in industry structure directly and in a way that is independent of variation in firm market shares, which would be the case if we were to use the leave-one-out HHI instead.

¹⁶We examine the robustness of this finding by instrumenting for endogenous concentration using the number of mergers and acquisitions in each local market, each year. This instrument is valid in that mergers among retail chains reflect broader economic trends than present in local markets (Arnold, 2021) and is also highly correlated with local HHI ($F = 2, 341.4$). Our point estimates remain negative, but not statistically different from zero in the preferred model (Model 5). See Table A1 in the Appendix.

Table 5 OLS Regression Estimates for NAICS 445: HHI and Wages, ACS Occupations

Dependent Variable	Log Real Weekly Wages				
	Model 1	Model 2	Model 3	Model 4	Model 5
log(HHI)	-0.0849*** (0.0136)	-0.0405*** (0.0126)	-0.0157*** (0.0062)	-0.0184*** (0.0063)	-0.0037 (0.0050)
<i>Specifications:</i>					
Controls	No	Yes	Yes	Yes	Yes
CZ Fixed Effects	No	No	Yes	Yes	Yes
Year Fixed Effects	No	No	No	Yes	Yes
CZ Trends	No	No	No	No	Yes
R ²	0.1151	0.2570	0.8878	0.8935	0.9145
RSS	1,355.17	1,137.30	171.71	163.02	130.84

All estimates with ACS, TDLinx and QCEW data for NAICS = 445 (Grocery Retailers) on a commuting-zone basis for the years 2005–2022. HHI is defined as the Hirschman–Herfindahl Index of employer concentration in each commuting zone, on an ACS-occupation basis, which forms an occupation definition of the relevant labor market. Model 1 consists of HHI only, Model 2 is Model 1 plus controls (population, income, age, and unemployment rate), Model 3 is Model 2 plus CZ fixed effects, Model 4 is Model 3 with year fixed effects, Model 5 is Model 4 with CZ-level trends. All standard errors are clustered at the CZ level. A single asterisk (*) indicates significance at a 10% level, ** at 5%, and *** at 1%. Dependent variable is log QCEW average weekly real wage at a commuting-zone level

Table 6 IV Estimates for NAICS 445: HHI and Wages, ACS Occupations

Dependent Variable	Log Real Weekly Wages (QCEW)				
	Model 1	Model 2	Model 3	Model 4	Model 5
log(HHI)	-0.0473*** (0.0201)	-0.0474 (0.0469)	0.1614 (0.2024)	0.1337 (0.0929)	0.0033 (0.0853)
<i>Specifications:</i>					
Controls	No	Yes	Yes	Yes	Yes
CZ Fixed Effects	No	No	Yes	Yes	Yes
Year Fixed Effects	No	No	No	Yes	Yes
CZ Trends	No	No	No	No	Yes
R ²	0.0925	0.2564	0.8372	0.8525	0.9145
RSS	1,389.79	1,138.18	271.01	225.84	130.89

All estimates with ACS, TDLinx and QCEW data for NAICS = 445 (Grocery Retailers) on a commuting-zone basis for the years 2005–2022. HHI is defined as the Hirschman–Herfindahl Index of employer concentration in each commuting zone, on an ACS-occupation basis, which forms an occupation definition of the relevant labor market. Model 1 consists of HHI only, Model 2 is Model 1 plus controls (population, income, age, and unemployment rate), Model 3 is Model 2 plus CZ fixed effects, Model 4 is Model 3 with year fixed effects, Model 5 is Model 4 with CZ-level trends. Instrument is defined as the leave-one-out inverse employer county by CZ and Year. All standard errors are clustered at the CZ level. A single asterisk (*) indicates significance at a 10% level, ** at 5%, and *** at 1%. Dependent variable is log QCEW average weekly real wage at a commuting-zone level

fact, our point estimates for the first model (Model 1, no controls or fixed effects) is very similar to theirs.

However, once we account for CZ-level controls, and fixed spatial and temporal effects (Model 4), the negative effect becomes very small (elasticity = 0.018). Finally, once we control for CZ-level trends the estimate is not significantly different from

zero. Again, there appears to be no relationship between concentration and wages in reduced-form models.

Our conclusion in Table 5 remains after we account for the endogeneity of labor-market concentration, with the use of both the leave-one-out instrumental variable (LOOIV) that we described above, and the number of mergers and acquisitions in each local market as in Arnold (2021) (see Table A1 in the appendix). In fact, in our preferred estimate in Table 6 (Model 5), the point estimate is again positive and not precisely estimated.¹⁷ After controlling for the endogeneity of market structure, and an occupation-definition of concentration, there is no significant relationship between HHI and wages.

In the appendix (Table A2) we examine an alternative analysis that uses CPS-ORG data in order to control for unionization, which shows a similar null effect of concentration on wages. Therefore, based on this occupation-based definition of concentration, we still find no relationship between concentration and wages in local labor markets. As we argue above, however, capturing the true effect of concentration on wage outcomes requires a structural model of how concentration affects bargaining power. In the next section, we connect concentration and bargaining power by using a structural search, match, and bargaining model.

5.2 Structural Bargaining Power Estimates

In Table 7 we first present and interpret our baseline structural estimates in order to establish the validity of our search, match, and bargaining power estimates, and then we allow bargaining power to vary with the level of local labor market concentration. Although our estimates in this model are still subject to the criticism that concentration is a weak measure of market power, these models provide the critical linkage between market structure and the exercise of market power that is absent in simple models of concentration and wage setting. In this regard, our models in this section follow the more recent macroeconomics literature in Berger et al. (2022) and Berger et al. (2024) in examining monopsony wage suppression with the use of the degree of bargaining power as our metric.

In Model 1 of Table 7, we estimate a base SMB model in which we do not include any measure of concentration in order to establish a baseline set of parameters that appear to characterize equilibrium bargaining among retail workers. In the case of Model 1, the structural parameters imply that matches are destroyed – or current workers are fired or laid off (δ) – at a rate of 18.31% per year, and that firms and workers create matches for those who are currently unemployed at a rate of only 3.32% per year. This finding suggests that there is very little recruitment out of the unemployed, so most new hires must be currently-employed individuals moving between firms, which is a common occurrence when wages are rising (Richards & Rutledge, 2023). Further, the parameters of the match-productivity function (ϕ and σ) imply an equilibrium match value of \$34.74 per hour which, when compared to the threshold match value that workers require to enter the workforce ($\phi^* = \$4.12/$

¹⁷ Estimates using the alternative, mergers and acquisitions, instrument are also not statistically different from zero.

Table 7 Structural Model of Concentration and Wages

Parameter	Model 1	Model 2	Model 3	Model 4
δ	0.1831*** (0.0001)	0.1464*** (0.0001)	0.1407*** (0.0001)	0.1133*** (0.0001)
τ	0.0332*** (0.0001)	0.0280*** (0.0001)	0.0252*** (0.0001)	0.0263*** (0.0001)
ϕ	2.9648*** (0.0001)	3.0067*** (0.0001)	3.0100*** (0.0001)	2.9507*** (0.0001)
σ	1.0800*** (0.0001)	1.1740*** (0.0001)	1.1197*** (0.0001)	1.3257*** (0.0001)
ϕ^*	4.1237*** (0.0014)	4.2713*** (0.0013)	4.1624*** (0.0022)	5.2290*** (0.0005)
λ	0.4891*** (0.0002)	0.4932*** (0.0002)	0.5096*** (0.0012)	0.4703*** (0.0002)
$\lambda(\text{HHI})$		-0.1164*** (0.0065)	0.0071 (0.0045)	-0.0629*** (0.0203)
LLF/N	18.1586	22.4359	11.0281	12.8731
N	425,404	425,404	425,404	425,404

Estimated with 2004 - 2022 American Community Survey (ACS, U.S. Census) data for workers in retailing occupations. Model 1 is the base search, match, and bargaining model with corrected (with demographic attributes) real wage data. Model 2 allows concentration to mediate the bargaining power parameter. Model 3 uses the TDLinx definition of concentration on an industry, instead of occupation, basis. Model 4 accounts for HHI endogeneity using a control-function approach and the leave-one-out IV from Tables 4 and 6. A single asterisk indicates significance at a 10% level, ** at 5%, and *** at 1%

hour) implies a surplus per worker of \$30.62 per hour.¹⁸ Workers and firms bargain over this surplus, and workers earn a $\lambda = 48.91\%$ share, according to our baseline bargaining power estimates. Relative to other estimates of bargaining power that is exercised by workers in the food system more generally (Richards & Rutledge, 2023), this estimate is quite high and suggests that workers in the retail food industry enjoy a relatively high degree of bargaining power – yet still give up nearly half of their earned surplus to employers.

In Model 2 of Table 7, we extend our base model results (Model 1) to account for the fact that bargaining power may vary across markets with different levels of labor-market concentration among the top employers. We estimate the same model as in Model 1, but add an interaction effect between the ACS definition of HHI and the bargaining power parameter. In this way, we account for the possibility that bargaining power – and hence wages – depends directly on concentration in a structural way.

In Model 2, the estimates in Table 7 show that all of the parameters change slightly with the addition of concentration as an argument of bargaining power: The estimate of job destruction (δ) falls to 14.64% per year, while the estimate of job creation (τ) falls to 2.8%. The estimated threshold match value rises to $\phi^* = \$4.27$ per hour, which suggests that workers require slightly more to move into a job, while the equilibrium productivity of a match is marginally higher at \$40.28 per hour.

¹⁸ Recall that match-productivity is distributed log-normal, so we calculate the value of a match as $f = \exp(\phi + 0.5\sigma^2)$.

Combining these estimates, we find that the equilibrium surplus is now \$36.01 per hour, with workers earning over $\lambda = 49.32\%$ of the equilibrium surplus. Most important for our purposes, we find that worker bargaining power falls significantly in the level of market concentration. Given that we calibrate the HHI index so that a value of $\text{HHI} = 0$ represents a perfectly competitive industry and $\text{HHI} = 1$ implies monopoly, an increase of HHI of 0.1 in our calibration (or moving from an HHI of 2, 500 to 3, 500 using the Federal Trade Commission calibration of HHI) is associated with a reduction in bargaining power from $\lambda = 49.32\%$ to $\lambda' = 37.68\%$: a reduction in the amount of surplus that is paid to workers of roughly \$1.05 per hour. At the mean level of wages in the data (\$9.04 per hour, see Table 1), concentration in the retail food industry is responsible for a 11.64% reduction in average wages.

In Model 4 of Table 7, we use the leave-one-out IV and a control-function approach to account for the potential endogeneity of HHI and find that an increase of HHI of 0.1 leads to a reduction in bargaining power from $\lambda = 47.03\%$ to $\lambda' = 40.74\%$. This translates to a 6.29% or \$0.57 per hour reduction in average wages.

While the results from both Models 2 and 4 of Table 7 suggest a relatively small decrease of hourly wages in concentrated labor markets, the reduction is economically significant for most workers in food retailing. Because union-coverage may be important for estimates of bargaining power, we examine data from a sample of workers from the Current Population Survey - Outgoing Rotation Group (CPS-ORG) and find a similar relationship between bargaining and labor-market concentration even in the presence of unionization, although union coverage appears likely to offset the negative effect of concentration (see Table A3 in the appendix).

Our findings are of practical and policy importance, and are remarkably similar to the findings in the rest of the concentration-and-wages literature (Azar et al., 2020, 2022; Qiu & Sojourner, 2023; Rinz, 2022) – despite the fact that we use an entirely different approach. Our results imply that mergers that result in a 1, 000-point increase in the Federal Trade Commission HHI measure can be expected to lead to substantial reductions in retail-grocery workers' wages. In the context of the proposed merger between Kroger and Albertsons in the fall of 2022, for example, many of the markets in which the two firms overlap were expected to experience a rise in HHI of this magnitude. Whereas traditional merger analysis would have taken only the price effect of a merger into account, our findings show that wages should also be taken into account.

6 Conclusions

Antitrust policy has a new focus on labor market outcomes: monopsony wage suppression. Empirical research that addresses the relationship between monopsony power and labor market outcomes tends to use reduced-form approaches that examine measures of labor-market structure on wages, employment, and sometimes job mobility.

In this paper, we argue that this focus on market structure ignores both the fundamental endogeneity of market concentration (Miller et al., 2022) and the behavioral gap between market structure and the actual exercise of market power. To that end,

we study the relationship between market structure, market power, and wages in the retail food industry; we use both reduced form and structural models of monopsony wage suppression.

We combine firm-level data from NielsenIQ's TDLinx data product with worker-level data from the ACS (U.S. Bureau of Census 2023). Because TDLinx contains data on outlet- and firm-level employment, we are able to construct commuting-zone (CZ) level measures of labor-market concentration; but we argue that measuring concentration at the industry level ignores the inherent fungibility of jobs across retail industries: Workers who take jobs in supermarkets can use their skills in many other industries, so labor markets are better described by occupations than industries.

We apply this insight by defining labor market concentration across all retail occupations in each CZ in the U.S., and use worker-level outcomes from the ACS and retail-industry wages from the Quarterly Census of Employment and Wages (BLS, U.S. Department of Labor, 2025) to estimate the effect of labor-market concentration on wages in both reduced-form and structural modeling frameworks.

Our structural model connects labor market concentration – our measure of the structure of local labor markets – to wage outcomes through a bargaining power mechanism. This insight is important, because concentration, *per se*, is only indirectly linked to labor market outcomes, since it is but one imperfection that may cause wages to differ from their marginal value products (Berger et al., 2024). We use a structural model of search, matching, and bargaining to connect market structure to the equilibrium bargaining power that is exercised by firms and employees in order to estimate the ultimate effect of labor-market concentration on wages.

Estimates from reduced-form models show that there is little or no statistical relationship between concentration and wages in local markets for retailing labor, after we control for a wide range of covariates and methods of addressing the clear endogeneity of market concentration. Although our estimates from models in the ACS data show a negative correlation between job-market concentration and wages that are similar to others in the literature, the relationship is not robust to controlling for the endogeneity of concentration. We interpret this finding as suggesting that many of the same factors that drive increasing structural concentration in the retail food industry – increasing competitiveness and the need to control costs – are also responsible for the relative decline in retailing wages over time. However, once we control for endogenous job formation and bargaining between retailers and their employees – and we allow concentration to moderate the level of bargaining power that is exercised by workers – we find that retailers do indeed have greater bargaining power in markets that are more concentrated. Consequently, we find that wages in these markets are lower than in markets that are less concentrated, all else constant.

Our structural bargaining model also addresses a key question in antitrust law with regard to the role of labor in Clayton Act merger cases. In the proposed Kroger-Albertsons merger, the Federal Trade Commission made an argument that allowing the merger would reduce the bargaining power of unionized workers. Because our model focuses specifically on the effect of concentration on bargaining power, and not on market power *per se*, our approach is exactly suited to answering the question that was posed by Judge Nelson in her December 2024 decision that rejected the merger: She states in her decision that neither the plaintiffs nor the defendants in the

Kroger–Albertsons case presented relevant HHIs or systematic economic modeling that could be used to guide future Clayton Act cases that may involve labor.

We do that here, and believe that we are the first to model formally the “new” consideration of labor markets in the 2023 Merger Guidelines.

Our findings are important as they provide a key link in the new approach to empirical antitrust analysis that has been missing. Reduced-form models of concentration and wages simply miss the point: Structure does not necessarily change wage outcomes; but bargaining power certainly does. Our findings provide an important step in the analysis of the effect of mergers and acquisitions on worker welfare, and the likely effect on wages in markets that are already highly concentrated. Empirical antitrust analysis of any proposed merger, therefore, needs to go the extra mile and connect any change in market concentration with the exercise of bargaining power.

The analysis in our paper contains inevitable weaknesses, many of which can be addressed through better data: First, matched employer–employee data remains the gold standard for labor-market analysis such as this. While there are some initial efforts in that regard in the U.S. that use data from the Longitudinal Employer–Household Dynamics (LEHD, U.S. Census Bureau) and Longitudinal Business Database (LBD) (Arnold, 2021), access to these data remains restricted, and merging the two remains uncertain at best. Matching employees to employers – as in Cahuc et al. (2006) or more recently in Berger et al. (2024), who use European data – represents the ideal in this literature.

Second, there are many ways to define occupations across different industries, so our reliance on ACS categorizations is imperfect and a weakness in our approach. Third, TDLinx is a census of retail operations, but does not include all employees in the retail food industry. To the extent that TDLinx misses some of the workers employed by major food retailers in the U.S., our concentration measures will contain some error.

Appendix A: Structural Model Details

In this appendix, we provide more details on the structural model of search, match, and bargaining that we outlined in the main text.

Following the standard approach in this literature (Flinn, 2006; Pissarides, 2000), our model assumes that match productivity (ϕ) is distributed across the sample of workers according to a distribution function – $f(\phi)$ – which is assumed to be log-normal and is determined by the production technology that is used by the firm.¹⁹ We follow Flinn (2006) in what follows, and assume that both firms and workers observe the productive value of the match – ϕ – and that match productivity is exogenously determined by the attributes of the employee, and by the quality of the match with

¹⁹ A reviewer questions whether match productivity should also depend on concentration. However, allowing employment surplus to depend on concentration assumes that retailers compete as Cournot oligopolists, while the empirical evidence holds that they not only compete as Bertrand rivals (Villas-Boas, 2007), but very aggressively as variable markups are only 5% on a multiple-category basis (Thomassen et al., 2017) even before allowing for fixed costs. Further, introducing concentration in this way would be ad hoc and inconsistent with the structure of the Flinn (2006) model.

a particular employer. The exogenous rate of job termination for employed workers is δ ; the exogenous rate of job contacts, or the rate at which jobs are created, for unemployed workers is τ ; and the discount rate is $\beta > 0$. Workers negotiate with an exogenously determined amount of bargaining power – $\lambda \in (0, 1)$ – which reflects the share of the match surplus they retain from the employment relationship. In our application of the Flinn (2006) model, we recognize that bargaining power (λ) is likely to be a function of the relative concentration of employers and employees in the labor market, based on recent insights on this fact from the literature (Arnold, 2021; Azar et al., 2020, 2022; Benmelech et al., 2022; Qiu & Sojourner, 2023; Rinz, 2022; Yeh et al., 2022); so we consider bargaining power as a structural function of concentration, or $\lambda = \lambda(h)$ where h is our measure of concentration, HHI_{ct} above.

In this model, labor is considered the only factor of production, so firms earn zero profit, and derive no value from participating in the labor market if they do not employ workers. The profit from employing a worker is the difference between their productivity value and the wage, or $\phi - w$, where w is the wage that is paid to employees. Employers and employees negotiate according to a Nash (1950) bargaining framework in which the strength of each player's bargaining position depends on the value of their next best alternative, or their "disagreement profit" in Nash bargaining terminology. From the worker's perspective, the disagreement profit is the value of staying unemployed, during which time they are assumed to continue searching for a job, so their next best alternative is the value of ongoing job search efforts, denoted by W_u . At a certain point, there is a threshold wage – the critical match value, ϕ^* – that determines whether an unemployed worker will accept a job offer. The critical match value is given by $\phi^* = \beta W_u$, and determines whether labor will be supplied, as all values of ϕ that meet or exceed ϕ^* will result in employment while those that are lower will not. Once a worker accepts a job offer, the value of employment depends on the wage: $W_e(w)$.

With these assumptions, the value to a worker of taking a job is the present value of their wage, plus the expected present value of reverting to unemployment, or:

$$W_e = \frac{w + \delta W_u}{\beta + \delta}, \quad (2)$$

where the "effective" discount rate includes a risk premium for the possibility of becoming unemployed. At the same time, the value of unemployed search is equal to the reservation wage (or the utility of consuming leisure, R) plus the expected present value of the surplus that is earned from taking a job at any wage greater than the critical match value:

$$\beta W_u = R + \frac{\lambda \tau}{\beta + \delta} \int_{\beta W_u} [\phi - \beta W_u] df(\phi), \quad (3)$$

where the expected value is over the entire distribution of possible match values above the reservation value.

From the firm's perspective, the value of employing a worker with productivity level ϕ is simply the present value of the amount of employment surplus, discounted at the same rate as the employee, or:

$$W_f = \frac{\phi - w}{\beta + \delta}, \quad (4)$$

where the wage is the observed hourly wage for a typical worker.

Once a match occurs, employers and employees bargain for wages at all values of $\phi \geq \phi^*$, that solve the generalized Nash bargaining problem:

$$w(\phi, W_u) = \arg \max_w [W_e(w) - W_u]^{\lambda(h)} \left[\frac{\phi - w}{\beta + \delta} \right]^{1-\lambda(h)}, \quad (5)$$

where $\lambda(h)$ measures the share of employment surplus that is earned by the employee, and $(1 - \lambda(h))$ is the share that is earned by the employer; recall that the disagreement profit for the firm is zero. As explained above, the key determinant of bargaining power is the degree of labor-market concentration, so $\lambda(h)$, and we test for the sign of $\lambda'(h)$ in our model. In the absence of any consideration for minimum wages, the wage that solves (5) is given by:

$$w(\phi, W_u) = \lambda(h)\phi + (1 - \lambda(h))\beta W_u, \quad (6)$$

where $\beta W_u = \phi^*$ is the threshold match value for the employee. However, minimum wages are an important feature of the low- and semi-skilled retail labor market (7.4% of our sample), so we follow Flinn (2006) in modifying the problem to allow explicitly for the imposition of minimum wages on market-driven wage bargaining.

Minimum wages constrain the wages that employers can pay. However, because employers earn some surplus on each employee – depending on the realized values of ϕ and $\lambda(h)$ – they have the ability to give up some surplus to hire workers with ϕ greater than the minimum wage (w_m) even though the minimum wage may be greater than the equilibrium wage suggested by (6). To see this more formally, assume that the value of unemployed search is now a function of the minimum wage, and solve for the threshold value of $\hat{\phi}$ that holds when $w = w_m$, or:

$$\hat{\phi}(w_m, W_u(w_m)) = \frac{w_m - (1 - \lambda(h))W_u(w_m)}{\lambda(h)},$$

so the minimum wage defines regions of the equilibrium match value that separate workers who are paid clearly above the minimum wage, those who are paid the minimum wage, and those who are not hired at all. Because minimum wages impose a discontinuity on the distribution of wages, the value of unemployed search becomes:

$$\beta W_u(w_m) = R + \frac{\tau}{\beta + \delta} \left\{ \int_{w_m}^{\hat{\phi}} [w_m - \beta W_u(w_m)] df(\phi) + \lambda(h) \int_{\hat{\phi}}^{\infty} [\phi(h) - \beta W_u(w_m)] df(\phi) \right\}, \quad (7)$$

which changes the solution for the equilibrium market wage. Substituting (7) back into the Nash bargaining problem in (5) and solving for the equilibrium wage distribution leads to:

$$g(w; W_u(w_m)) = \begin{cases} [f'(\hat{\phi}(w, W_u(w_m)))/\lambda(h)f(w_m), & w > w_m \\ [f(w_m) - f(\hat{\phi}(w, W_u(w_m)))]/f(w_m), & w = w_m \\ 0, & w < w_m \end{cases}, \quad (8)$$

for workers who are paid above the minimum wage, at the minimum wage, or who are not hired, respectively. We use the equilibrium wage distribution in (8) to derive several hypotheses with regard to the likely effect of labor-market concentration on equilibrium wages, employment, and the duration of unemployment.

We estimate the model with data on real observed wages (w_i), adjusted for demographic (age, gender, education, marital status, and income), job, worker classification, year and state effects, and the amount of time spent unemployed during the past year (t_i) for a repeated cross-section of some $N = 426,000$ worker-year observations. All of our structural data describe employees in the U.S. who work in occupations relevant to the retailing industry. We describe our data in more detail below but find that it is sufficient to identify all of the parameters of the wage distribution above, including the Nash bargaining parameter, $\lambda(h)$, that determines the share of employment surplus that is earned by employees, and by firms. In this section, we derive the log-likelihood function that was developed by Flinn (2006) that is used to estimate the parameters of search, matching, and bargaining models such as ours. We then explain how we test the hypotheses that we derived above with regard to the effect of labor-market concentration on wages, productivity, and job duration.

Because minimum-wage employment is common in retailing, we follow Flinn (2006) and break the likelihood function into three parts: (1) the probability that a worker is unemployed for a duration of t weeks; (2) the probability that a worker is employed and paid a wage that is equal to the minimum wage; and (3) the probability that a worker is employed and paid more than the minimum wage.

For the first component of the likelihood function, we have to assume a distribution for unemployment durations, in general. In this regard, like Flinn (2006) we follow common practice and assume that the distribution of the population duration function is negative exponential, so we write the individual-density of job duration as the mean, or:

$$pr(t|u) = \tau f(w_m) \exp(-\tau f(w_m)t), \quad (9)$$

where the parameters and minimum wage variable are as defined above. We use the parametric rate of job destruction to infer that the probability of a worker's becoming unemployed during the year is:

$$pr(u) = \frac{\delta}{\delta + \tau f(w_m)}. \quad (10)$$

Multiplying the conditional probability of observing a spell of length t , given that the worker is unemployed, by the marginal probability of becoming unemployed gives the joint probability of observing an employee becoming unemployed for a period of length t , or:

$$pr(t, u) = \frac{\delta \tau f(w_m) \exp(-\tau f(w_m)t)}{\delta + \tau f(w_m)}; \quad (11)$$

and we assume f is log-normal, with parameters μ for the mean and σ for the standard deviation.

Second, we derive the likelihood-component that captures the probability that a worker is employed and paid the minimum wage. Recall that the primary implication of a minimum wage regime is to constrain the range of equilibrium productivity values to those that lie above the minimum wage. With this in mind, the likelihood contribution from minimum-wage employees is given by:

$$pr(w = w_m, e) = \frac{\tau \left[f(w_m) - f\left(\frac{w_m - (1 - \lambda(h))\beta W_u(w_m)}{\lambda(h)}\right) \right]}{\delta + \tau f(w_m)}, \quad (12)$$

which represents the probability a worker is employed (e) but is paid the minimum wage, so the firm is willing to give up some surplus in order to hire a worker that still produces value greater than the level of the minimum wage.

A third segment of workers are employed and paid above the minimum wage. For these workers, the minimum wage is still relevant, however, as it remains an element of the value of unemployed search, which, in turn, determines the threshold match value for employment. The probability of observing a wage w for an employed worker, therefore, is the product of the conditional probability of observing a worker being paid above the minimum wage – conditional on being employed – and the probability of observing a particular wage above the minimum. The second element of this expression, therefore, is given by:

$$pr(w|w > w_m, e) = \frac{\frac{1}{\lambda(h)} f' \left(\frac{w - (1 - \lambda(h))\beta W_u(w_m)}{\lambda(h)} \right)}{f \left(\frac{w_m - (1 - \lambda(h))\beta W_u(w_m)}{\lambda(h)} \right)}, \quad (13)$$

as the wage has to exceed the match-minimum of $\frac{w_m - (1 - \lambda(h))\beta W_u(w_m)}{\lambda(h)}$. Meanwhile, the conditional probability that a worker's wage is above the minimum, conditional on employment, is:

$$pr(w > w_m | e) = \frac{f\left(\frac{w_m - (1 - \lambda(h))\beta W_u(w_m)}{\lambda(h)}\right)}{f(w_m)}. \quad (14)$$

Multiplying these two expressions together provides the joint probability, and likelihood contribution, of observing a wage w that is above the minimum w_m for an employed worker:

$$f(w, w > w_m, e) = \frac{\frac{\tau}{\lambda(h)} f'\left(\frac{w - (1 - \lambda(h))\beta W_u(w_m)}{\lambda(h)}\right)}{\delta + \tau f(w_m)}. \quad (15)$$

Combining all three elements, taking logs, and summing over all individuals in the data set provides a likelihood function that recovers all of the parameters of interest:

$$\begin{aligned} LLF = & [\ln(\tau) - \ln(\delta + \tau f(w_m))] + d_U [\ln(\delta) + \ln f(w_m)] \\ & - \tau f(w_m) d_U t_i + d_M \ln \left(f(w_m) - f\left(\frac{w_m - (1 - \lambda(h))\phi^*}{\lambda(h)}\right) \right) \\ & - d_H \ln(\lambda(h)) + d_H \ln \left(f'\left(\frac{w_i - (1 - \lambda(h))\phi^*}{\lambda(h)}\right) \right), \end{aligned} \quad (16)$$

where d_U is a binary indicator of whether the worker is unemployed ($d_u = 1$) or employed ($d_u = 0$), d_M is a binary indicator that the worker is paid the minimum wage, and d_H is a binary indicator that the worker is paid above the minimum wage. With this likelihood function, we estimate the implicit minimum wage as $\phi^* = \beta W_u(w_m)$, which allows for a source of exogenous variation to identify the key primitive of the model.

Appendix B: Robustness Checks

In this appendix, we present the estimates from estimating the models in Table 4 in the main text using a commuting zone-level count of mergers and acquisitions as an instrument for the retail HHI variable in the empirical model (Table A1, Arnold 2021). We also examine the effect of accounting for unionization in both our reduced-form wage and search, match, and bargaining models in Tables A2 and A3, respectively. We first present the results of our alternative-IV estimation results in Table A1. The results in this table show that none of the HHI estimates are statistically different from zero, but the point estimates are negative, which supports the findings from our maintained ACS-QCEW model in the main paper.

Table 8 IV Model Estimates for NAICS 445: HHI and Wages at the CZ Level

Dependent Variable	Log Real Weekly Wages				
	Model 1	Model 2	Model 3	Model 4	Model 5
log(HHI)	-0.6764***	-0.0346	0.0450***	-0.0169	-0.0022

Table 8 IV Model Estimates for NAICS 445: HHI and Wages at the CZ Level

Dependent Variable	Log Real Weekly Wages				
	Model 1	Model 2	Model 3	Model 4	Model 5
	(0.0685)	(0.0316)	(0.0132)	(0.0158)	(0.0149)
<i>Specifications:</i>					
Controls	No	Yes	Yes	Yes	Yes
CZ Fixed Effects	No	No	Yes	Yes	Yes
Year Fixed Effects	No	No	No	Yes	Yes
CZ Trends	No	No	No	No	Yes
R^2	0.0459	0.2799	0.8162	0.8222	0.8616
RSS	9,613.12	7,242.49	1,848.67	1,787.65	1,391.97

All estimates with TDLinx and QCEW data for NAICS = 445 (Grocery Retailers) on a commuting-zone basis for the years 2005–2022. HHI is defined as the Hirschman–Herfindahl Index of employer concentration in each commuting zone, which forms an "industry" definition of the relevant labor market. Model 1 consists of HHI only, Model 2 is Model 1 plus controls (population, income, age, and unemployment rate), Model 3 is Model 2 plus CZ fixed effects, Model 4 is Model 3 with year fixed effects, Model 5 is Model 4 with CZ-level trends. IV models use the log of the number of mergers and acquisitions in the retailing industry in each CZ-year observation (Arnold, 2021). All standard errors are in parenthesis and clustered at the CZ level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

We next present our findings from applying a subset of our reduced-form concentration-and-wage models in Tables 3–6 and our search, match, and bargaining model in Table 7 to data from the Current Population Survey - Outgoing Rotation Group (CPS-ORG, U.S. Bureau of Census) in order to examine the effect of unionization on wages and bargaining power, respectively. We use data from the CPS-ORG for this purpose because there are no consistent data on union membership either by retail ownership group or by geography. Even within banners owned by retailers that are generally unionized, union coverage varies greatly from market to market and from occupation to occupation. Therefore, the only tractable way to study the effect of union membership on wages and bargaining outcomes is through an individual-level data set such as the CPS. Our CPS-ORG sample mirrors that from the sample of ACS and CPS-ASEC retail workers described above. Specifically, we choose a sample of workers in both the food retailing and adjacent industries that perform the same tasks as those that are commonly performed in food retailing over the same 2005–2022 time period. Only ORG respondents report union membership, so the summary CPS-ASEC data in Table 2 are not appropriate for this purpose. We include not only workers who actively belong to unions, but also non-union-member workers who are covered by union collective bargaining agreements.

With these data, the estimates in Table A2 are from regression models that are identical to those that we used to generate Tables 3–6, but are estimated at the individual level, with state, year, and occupation fixed effects. For the models that address HHI endogeneity (Models 4 and 5), we use the same instrumental variable (leave one out inverse number of employers) in order to maintain comparability with the estimates in Tables 3–6. Based on these estimates, the insights from Table A2 are qualitatively similar to those that are reported in the main text: Concentration has no measurable effect on wages after appropriately controlling for the endogeneity of concentration and state-level trends. Interestingly, the Union effect in Table A2 almost exactly counteracts the Citizenship effect, where we define Citizenship = 1 for a non-citizen

in the CPS-ORG data, and this finding is robust across all models that include Occupation fixed effects (Models 2–5).

Table 9 Estimates for Retail Workers, CPS ORG Wages and Occupation HHI

Dependent Variable	Log Real Hourly Wages				
	Model 1	Model 2	Model 3	Model 4	Model 5
<i>Main Variables:</i>					
Citizenship	-0.1382*** (0.0344)	-0.0959*** (0.0326)	-0.0929*** (0.0335)	-0.0979*** (0.0315)	-0.0943*** (0.0327)
Union	0.0236 (0.0287)	0.0987*** (0.0277)	0.0962*** (0.0279)	0.0967*** (0.0285)	0.0956*** (0.0279)
log(HHI)	0.0405** (0.0229)	0.0427*** (0.0220)	0.0176 (0.0233)	0.1172 (0.1316)	0.1619 (0.1470)
Constant	0.4089*** (0.1542)	0.2777 (0.3977)	0.1231 (0.4173)	0.6469 (0.8949)	0.8777 (0.9888)
<i>Specifications:</i>					
Controls	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Occupation Controls	No	Yes	Yes	Yes	Yes
State Trends	No	No	Yes	No	Yes
IV for HHI	No	No	No	Yes	Yes
R^2	0.2906	0.3777	0.3825	0.3768	0.3811
RSS	3,047.47	2,673.13	2,652.80	2,625.69	2,607.54

All estimates with CPS-ORG real wages for workers in food retailing on an individual-level basis for years 2005–2022. HHI is defined as the Hirschman–Herfindahl Index of employer concentration in each state, on an ACS-occupation basis, which forms an occupation-based definition of the relevant labor market. Model 1 consists of state and year effects and controls (age, education, marital status, education). Model 2 adds controls for occupation, industry and worker classification, Model 3 is Model 2 plus state-level trends, Model 4 is Model 2 with IV (leave one out inverse number of employers) for HHI, Model 5 is Model 4 with state-level trends. F-statistic for IV regression is 154.67 so instrument is not weak. All standard errors are in parenthesis and clustered at the state level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

We then examine the effect of unionization on bargaining power with the use of the Flinn (2006) search, match, and bargaining model in the CPS-ORG sample from 2005–2022. Our sample from these data is small relative to the ACS sample that we used in the main text because only a subset of the CPS-ORG respondents provided data on the number of weeks unemployed, their annual wage income, and the number of hours worked – each of which is necessary to estimate our structural model of bargaining. Using the same model as we apply to the ACS data in the main text (except for the fact that we normalize $\sigma = 1$ for identification in the smaller sample) we find similar results in Table A3: Bargaining power is lower by roughly 4 percentage points (6 percentage points in the ACS data) for each point of HHI index, where $HHI = 0$ is a perfectly competitive labor market and $HHI = 1$ is a monopsony employer. In this table, however, we show that union coverage more than completely offsets the concentration effect in the sense that a union worker would be able completely to address any bargaining differences that may exist between an employer in a perfectly competitive industry relative to a perfect monopsonist. Because the HHI effect in Model 4, with unionization, is slightly below the equivalent estimate in Model 2,

without unionization, we can infer that the bargaining power estimate in Table 7 is likely to be similarly overestimated due to our inability to control for unionization in the ACS-based model. Nonetheless, these estimates provide some support for the robustness of our bargaining power and concentration estimates in Table 7.

Table 10 Structural Model of Concentration, Wages, 3and Unionization

Parameter	Model 1	Model 2	Model 3	Model 4
δ	0.0150*** (0.0021)	0.1377*** (0.0104)	0.0627*** (0.0084)	0.0111*** (0.0045)
τ	0.0022*** (0.0002)	0.0269*** (0.0023)	0.0026*** (0.0003)	0.0027*** (0.0002)
μ	1.2814*** (0.0316)	2.1132*** (0.0236)	1.9536*** (0.0219)	1.0743*** (0.0595)
ϕ	4.2500*** (0.0887)	5.6812*** (0.0912)	4.8605*** (0.0566)	6.6628*** (0.1495)
λ	0.6153*** (0.0046)	0.6218*** (0.0036)	0.5032*** (0.0029)	0.6742*** (0.0047)
$\lambda(\text{HHI})$		-0.0641*** (0.0059)	-0.0861*** (0.0001)	-0.0428*** (0.0097)
$\lambda(\text{Union})$			0.0323*** (0.0007)	0.0626*** (0.0016)
Control Function				-0.0459*** (0.0152)
AIC/N	6.9870	10.0310	11.6382	8.7911
N	7,541	7,541	7,541	7,541

Estimated with 2005 - 2022 Current Population Survey - Outgoing Rotation Group (CPS-ORG, U.S. Census) data for workers in retail occupations. Model 1 is the base search, match, and bargaining model with corrected (demographic and job attributes) real wage data. Model 2 allows concentration to mediate the bargaining power parameter. Model 3 introduces the effect of union coverage on bargaining power. Model 4 accounts for HHI endogeneity using a control-function approach and the leave-one-out IV from Tables 4 and 6. Note that σ is normalized to 1.0 for identification in this model, and the Control variable represents the control-function estimate. Standard errors are in parenthesis. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Author Contributions T.R. and U.P. conducted the analysis and wrote most of the manuscript, K.M. contributed to writing the manuscript, and helped organize the data and design the empirical analysis.

Data Availability The raw TDLinx data are protected and are not available due to data privacy laws. The processed data sets can be obtained through a research partnership with ERS-USDA and NielsenIQ. All of the American Community Survey and Current Population Survey data are publicly available through IPUMS (<https://www.ipums.org/>).

Declarations

Competing interests The authors declare no competing interests.

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